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TRW Automotive

Slip Control Software Design

Document ID	:	ARM-DDD-Arm_Whl_Control
Title	:	ARM-DDD-Arm_Whl_Control
Summary	:	Description of subsystem architecture and interface
Scope	:	Application: • arm_whl_control{.c, _priv.h, _sub.h} Calibration:
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Release Date	Author	Reviewer	Approved by
	A. Hoehn		

REFERENCES

Ref.	Document	Title	Ver.	Location

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1.0	30/05/2011	B. Etges		Initial Version
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1 ABSTRACT

This module transforms the ARM pressure request for front and rear axle to a pressure and slip command for each wheel.

2 ACRONYMS

ARM Active Roll Mitigation

ROI Roll Over Index

WSR Wheel Slip Regulator

Prim Primary

Sec Secondary

WTP Wheel Transition Pressure

YSC Yaw Stability Control

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

3 OVERVIEW

This document describes the purpose and behavior of Arm_Whl_Control.

3.1 DESIGN GOAL

Active Roll Mitigation (ARM) and AyLimiter pressure and slip commands are arbitrated and distributed to the specific wheels. In addition special functions like prefill and ArmOverride are able to command individual pressure commands for each wheel.

3.2 GENERAL REQUIREMENTS

1. ARM should prevent dynamic rollover.
2. ARM should prevent static rollover.
3. ARM should apply sufficient brake pressure to the selected wheels in order to reduce the kinetic energy.

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3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when ARM is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

None.

4 DETAILED DESIGN

This chapter outlines the detailed design of Arm_Whl_Control.

4.1 ALGORITHM DESCRIPTION

4.1.1 ArmWhlControl

The command arbitration between ARM and AyLimiter is performed by using the maximum of each brake pressure and wheel slip request. The system pressure is added to the brake pressure request to consider the master cylinder pressure. ARM slip command is calculated based on the ARM pressure command where as AyLimiter request always the same slip command. The pressure requests for front and rear axle are arbitrated separately.

The brake torque gradient is determined by AyLimitier brake torque gradient. Due to the fact, that the arbitration in ArmWhlControl uses the maximum of Arm and AyLimiter, a simultaneous control of Arm and AyLimiter is possible. The compiler switch AYLIM_ARM_CTRL_SAME_TIME enables a simultaneous control possibility.

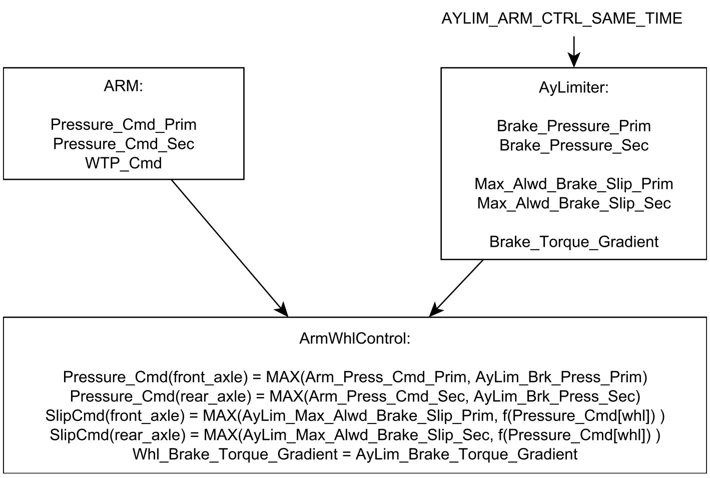

Figure 4.1: Pressure and Slip Arbitration

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ArmWhlControl contains some more special modes besides ArmOverride and ArmPrefill.

Arm_Bypass_Slip_Control

Arm_Bypass_Slip_Control can disable the Wheel Slip Regulator (WSR) in Yaw Stability Control (YSC). It is intended to be used during the ARM wheel transition, when the new ARM wheel does not yet have enough wheel load. In such a case, the wheel is likely to lock and the WSR will dump the pressure. However sometimes it is more important to have a very fast pressure build up than to unlock the wheel.

Arm_Request_Wsr_Hold

Arm_Request_Wsr_Hold disables pressure dumps of the YSC WSR. It should disable WSR pressure dumps in the second lane of an ARM maneuver. Wsr_Hold is reset, when the lateral acceleration has the first peak, which means the wheel has maximum wheel load.

Wheel Transition Pressure

Another special mode is the Wheel Transition Pressure (WTP). If WTP is active, the pressure will be requested on the opposite ARM wheel. This pressure overlap is only performed on the front axle to avoid four wheel braking which would result in a corrupted vehicle reference speed.

4.1.2 ArmSlipCommand

The function calculates for each wheel the Arm slip command based on the pressure command. A TwoDimensional Lookup table is used to determine the corresponding slip command. This command is further limited between two trimable thresholds.

4.1.3 ArmOverride

The function ArmOverride can override the YSC pressure and disable the YSC intervention at the front axle in order to enable the cross talk effect. In vehicles with front / rear split brake circuits this function is used to improve the pressure build up rate. Since the brake pressure at the previous side is transferred to the opposite side, the pressure build up rate increased. Therefore only one wheel at the front axle is allowed to be active, and the wheel change has to be performed within one loop, without any transition. The ArmOverride flag is set to disable YSC at the previous wheel, when Arm performs the wheel transition.

4.1.4 ArmPrefill

The function ArmPrefill is intended for a prefill on the opposite ARM wheel, in order to improve the pressure build up for a wheel transition.
When ARM is active and the brake pressure in the wheel, that is in ARM control, exceedes a threshold then the prefill is applied to the opposite wheel of the front axle. ArmPrefill is reset, after a certain duration, when the vehicle speed falls below a threshold, when the driver is braking or when the ArmFlag is reset.

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5 PROVIDED TASKS

5.1 ArmWhlControl

Description	ArmWhlControl
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void ArmState (void)

6 PROVIDED SERVICES

6.0.1 Private Services

- ArmSlipCommand
- ArmPrefill
- ArmOverride

7 CONFIGURATION OPTIONS

Name	Type	Range	Description
ENABLE_ARM_OVERRIDE	BOOLEAN	[0...1]	En-/Disable ARM overrides YSC command after wheel transition, to enable cross talk effect and improve pressure build up rate, in vehicles with front / rear split brake circuit configuration.
ENABLE_ARM_PREFILL	BOOLEAN	[0...1]	En-/Disable ARM prefill at the side of the vehicle, where ARM is not active.

8 CONFIGURATION PROCEDURES

Depending on project requirements and the roll over propensity of the vehicle the configurations has to be adjusted once before start of project. Changes during development phase of the project can be applied but should be carefully reviewed with the subsystem responsible Teamlead.

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9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

10.1 Wheel: Pressure Requests

The pressure requests are transformed to wheel level. Each request is limited between **PRESSURE_CMD_MIN** (robustness) and **PRESSURE_CMD_MAX** (safety). It is advised to kep the core trimming as long as there are no differing requirements.

It is possible to avoid pressure request on the rear axle if **ENABLE_ARM_REAR_WHL** is set to zero.

ARM_MAX_BRAKE_TORQUE_GRADIENT is used to pass a pressure gradient to the actuator. Since ARM relies on high pressure build up rates it is recommended to keep the core trimming.

10.2 Wheel: Slip Requests

Wheel slip requests are directly derived from the wheel specific pressure requests. They can be adjusted by **PRIM_SLIP_CMD_TBL** and **SEC_SLIP_CMD_TBL**. Increase the slip to avoid WSR pressure dumps. One can lower them to avoid wheel locks.

For WTP there is a constant parameter **WTP_SLIP_CMD**. Since the WTP wheel has low wheel load, the wheel slip can change fast.

11 APPENDIX